EXPERIMENT 8: NETFLIX CONTENT DATA ANALYSIS AND VISUALIZATION

AIM:

To analyze the Netflix content dataset using Python and Pandas, explore movies and TV shows, examine ratings, countries, release years, content growth, durations, genres and directors, and visualize content patterns and genre relationships.

ALGORITHM:

1. Import the required Python libraries and load the Netflix dataset into a Pandas DataFrame.

2. Explore the dataset by displaying its shape, columns, data types, and summary information.

3. Compare the number of Movies and TV Shows and visualize the distribution of content ratings.

4. Identify the top countries contributing Netflix content and analyze content by release year.

5. Extract the year content was added and visualize Netflix content growth over time and by month.

6. Analyze movie durations and the number of seasons for TV shows.

7. Examine the most common genres and top directors.

8. Compare ratings for Movies and TV Shows using a cross-tabulation and bar chart.

9. Build a genre co-occurrence table and visualize correlations between genres using a heatmap.

```python
# EXERCISE-7 Muthuraj R (241801176)
# STEP 1 LOAD NETFLIX DATASET
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset from the selected file path
df = pd.read_csv("/content/netflix_titles.csv")
print("Dataset Loaded Successfully!")
display(df.head())
```

```
Dataset Loaded Successfully!
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020.0	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021.0	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021.0	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021.0	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021.0	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

```python
# STEP 2 EXPLORE THE DATASET
# Muthuraj R (241801176)
print("Shape:", df.shape)
print("\nColumns:")
print(df.columns)
print("\nDataset Information:")
df.info()
print("\nStatistical Description:")
print(df.describe(include="all"))
print("\nMissing Values:")
print(df.isnull().sum())
```

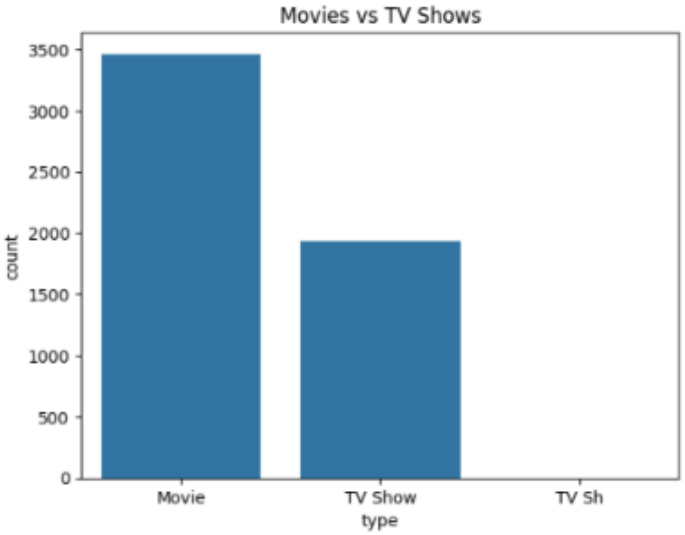
```
Shape: (5398, 12)

Columns:
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
       'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5398 entries, 0 to 5397
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       5398 non-null   object
 1   type          5398 non-null   object
 2   title         5397 non-null   object
 3   director      3515 non-null   object
 4   cast          4903 non-null   object
 5   country       4735 non-null   object
 6   date_added    5397 non-null   object
 7   release_year  5397 non-null   float64
 8   rating        5397 non-null   object
 9   duration      5397 non-null   object
 10  listed_in     5397 non-null   object
 11  description   5397 non-null   object
dtypes: float64(1), object(11)
memory usage: 506.2+ KB
```

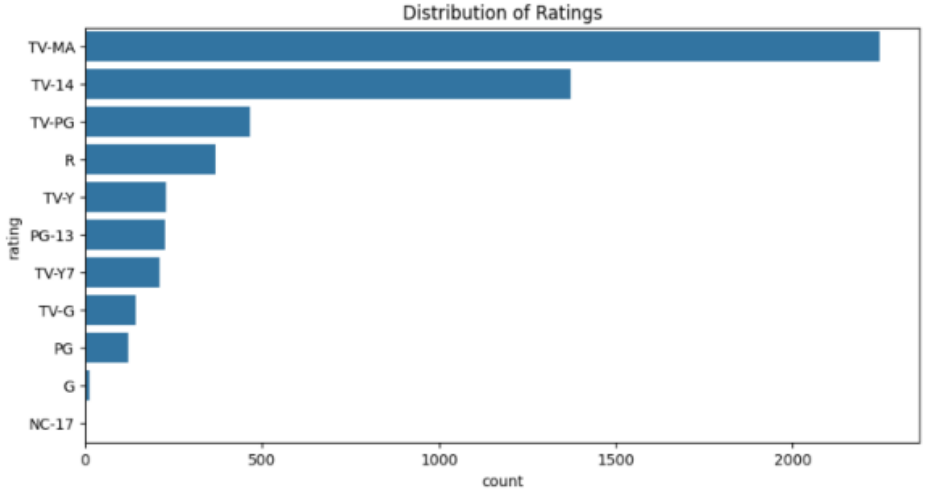
```python
# STEP 3(a) MOVIES VS TV SHOWS
# Muthuraj R (241801176)
print(df["type"].value_counts())
sns.countplot(data=df, x="type")
plt.title("Movies vs TV Shows")
plt.show()
```

```
type
Movie      3464
TV Show    1933
TV Sh         1
Name: count, dtype: int64
```



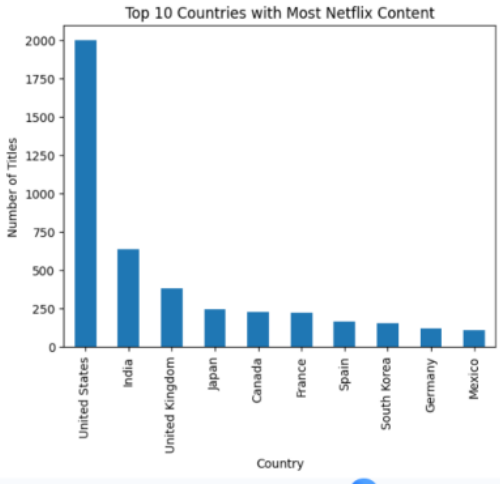

```python
# STEP 3(b) RATINGS DISTRIBUTION
# Muthuraj R (241801176)
print(df["rating"].value_counts())
plt.figure(figsize=(10, 5))
sns.countplot(data=df, y="rating", order=df["rating"].value_counts().index)
plt.title("Distribution of Ratings")
plt.show()
```

```
rating
TV-MA     2246
TV-14     1373
TV-PG      465
R          367
TV-Y       230
PG-13      226
TV-Y7      209
TV-G       144
PG         123
G           13
NC-17        1
Name: count, dtype: int64
```



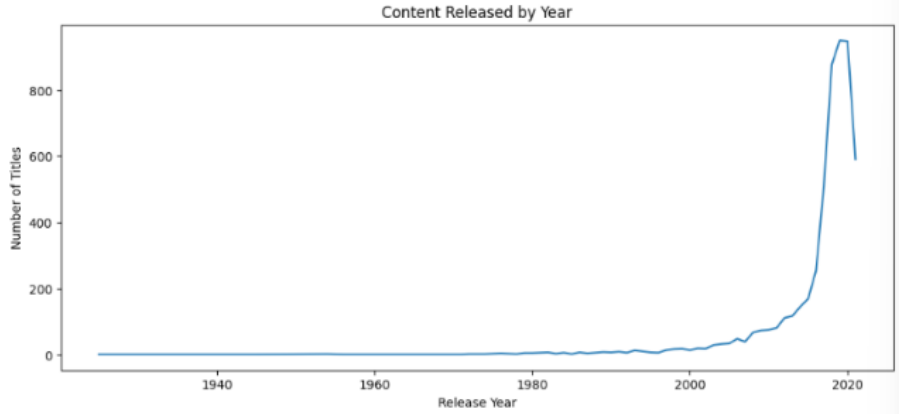
```python
# STEP 3(c) TOP COUNTRIES
# Muthuraj R (241801176)
countries = df["country"].dropna().str.split(", ").explode()
print(countries.value_counts().head(10))
countries.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Countries with Most Netflix Content")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.show()
```

```
country
United States     1999
India              638
United Kingdom     383
Japan              244
Canada             230
France             225
Spain              166
South Korea        158
Germany            124
Mexico             111
Name: count, dtype: int64
```



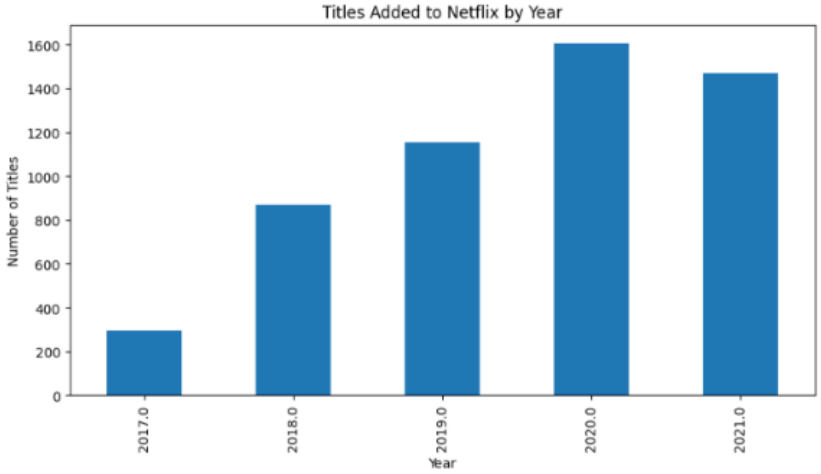

```python
# STEP 4(a) RELEASE YEAR ANALYSIS
# Muthuraj R (241801176)
print(df["release_year"].value_counts().head(10))
df["release_year"].value_counts().sort_index().plot(figsize=(12, 5))
plt.title("Content Released by Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()
```

```
release_year
2019.0    950
2020.0    948
2018.0    877
2021.0    591
2017.0    511
2016.0    254
2015.0    169
2014.0    145
2013.0    117
2012.0    111
Name: count, dtype: int64
```

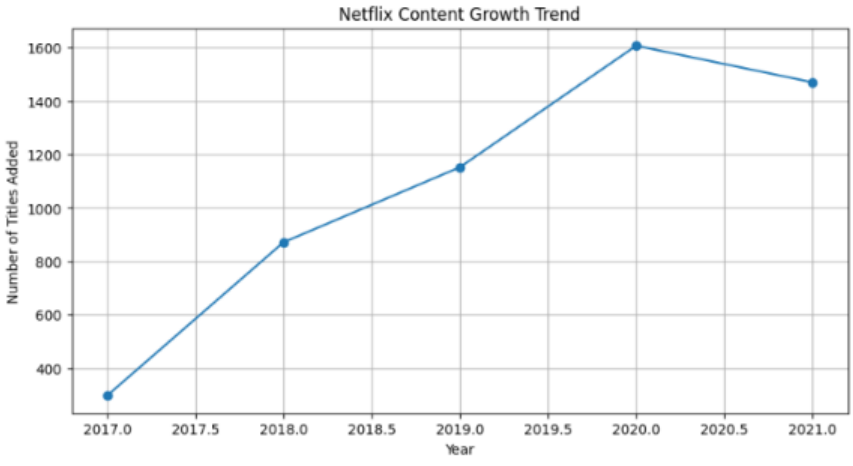


```python
# STEP 4(b) DATE ADDED YEAR
# Muthuraj R (241801176)
df["date_added"] = pd.to_datetime(df["date_added"], errors="coerce")
added_year = df["date_added"].dt.year.value_counts().sort_index()
print(added_year)
added_year.plot(kind="bar", figsize=(10, 5))
plt.title("Titles Added to Netflix by Year")
plt.xlabel("Year")
plt.ylabel("Number of Titles")
plt.show()
```

```
date_added
2017.0     297
2018.0     871
2019.0    1152
2020.0    1607
2021.0    1470
Name: count, dtype: int64
```

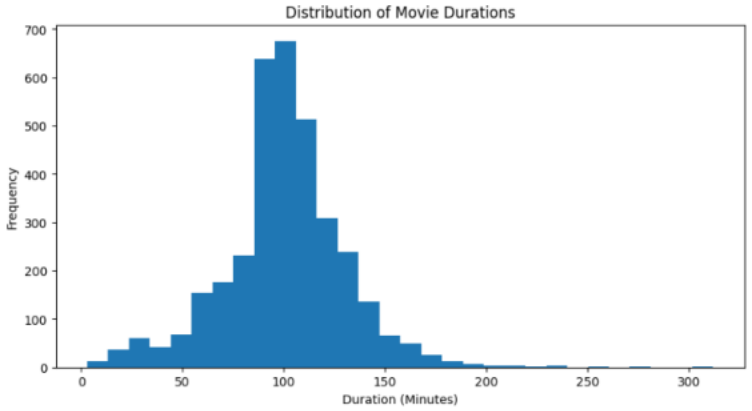



```python
# STEP 4(c) CONTENT GROWTH TREND
# Muthuraj R (241801176)
growth = df["date_added"].dt.year.value_counts().sort_index()
growth.plot(kind="line", marker="o", figsize=(10, 5))
plt.title("Netflix Content Growth Trend")
plt.xlabel("Year")
plt.ylabel("Number of Titles Added")
plt.grid()
plt.show()
```



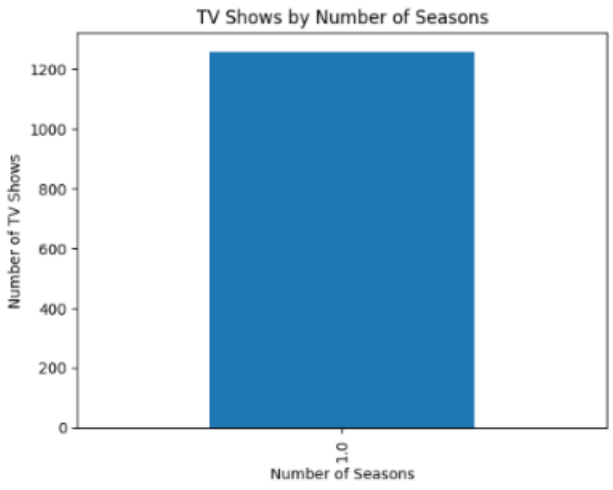
```python
# STEP 5(a) MOVIE DURATION DISTRIBUTION
movies = df[df["type"] == "Movie"].copy()
movies["duration_min"] = movies["duration"].str.replace(" min", "", regex=False).astype(float)
print(movies["duration_min"].describe())
movies["duration_min"].plot(kind="hist", bins=30, figsize=(10, 5))
plt.title("Distribution of Movie Durations")
plt.xlabel("Duration (Minutes)")
plt.show()
```

```
count    3464.000000
mean      100.387991
std        29.968087
min         3.000000
25%        87.000000
50%       100.000000
75%       116.000000
max       312.000000
Name: duration_min, dtype: float64
```



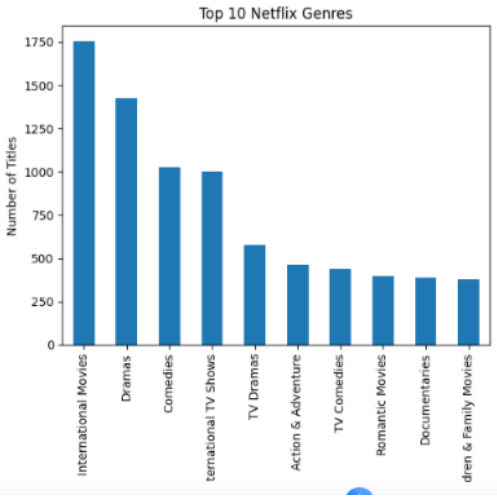

```python
# STEP 5(b) TV SHOW SEASONS
# Muthuraj R (241801176)
tv = df[df["type"] == "TV Show"].copy()
tv["seasons"] = pd.to_numeric(
    tv["duration"].str.replace(" Season", "", regex=False).str.replace(" Seasons", "", regex=False),
    errors="coerce"
)
print(tv["seasons"].value_counts().sort_index())
tv["seasons"].value_counts().sort_index().plot(kind="bar")
plt.title("TV Shows by Number of Seasons")
plt.xlabel("Number of Seasons")
plt.ylabel("Number of TV Shows")
plt.show()
```

```
seasons
1.0    1258
Name: count, dtype: int64
```



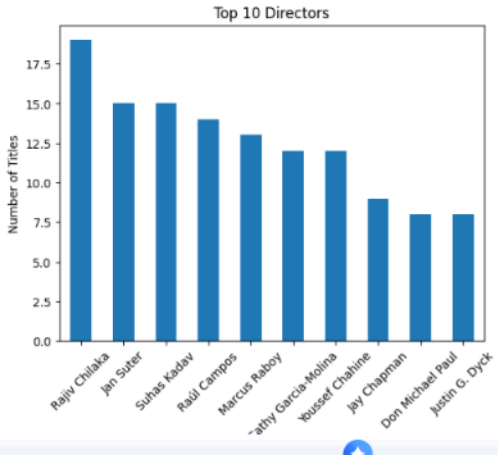
```python
# STEP 6(a) TOP GENRES
genres = df["listed_in"].dropna().str.split(", ").explode()
print(genres.value_counts().head(10))
genres.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Netflix Genres")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")
plt.show()
```

```
listed_in
International Movies        1755
Dramas                      1426
Comedies                    1025
International TV Shows      1000
TV Dramas                    576
Action & Adventure           464
TV Comedies                  436
Romantic Movies              398
Documentaries                389
Children & Family Movies     377
Name: count, dtype: int64
```



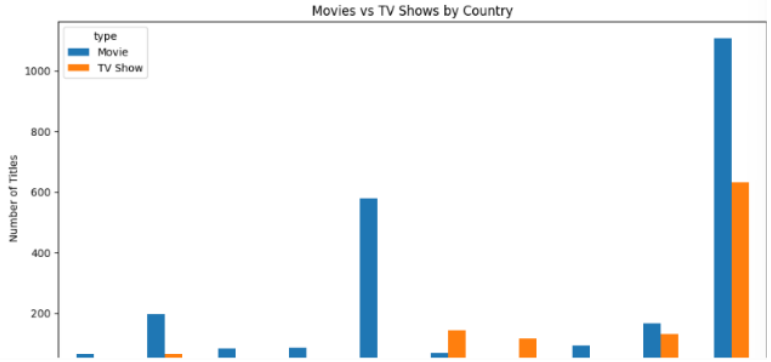

```python
# STEP 6(b) TOP DIRECTORS
# Muthuraj R (241801176)
directors = df["director"].dropna().str.split(", ").explode()
print(directors.value_counts().head(10))
directors.value_counts().head(10).plot(kind="bar")
plt.title("Top 10 Directors")
plt.xlabel("Director")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

```
director
Rajiv Chilaka          19
Jan Suter              15
Suhas Kadav            15
Raúl Campos            14
Marcus Raboy           13
Cathy Garcia-Molina    12
Youssef Chahine        12
Jay Chapman             9
Don Michael Paul        8
Justin G. Dyck          8
Name: count, dtype: int64
```



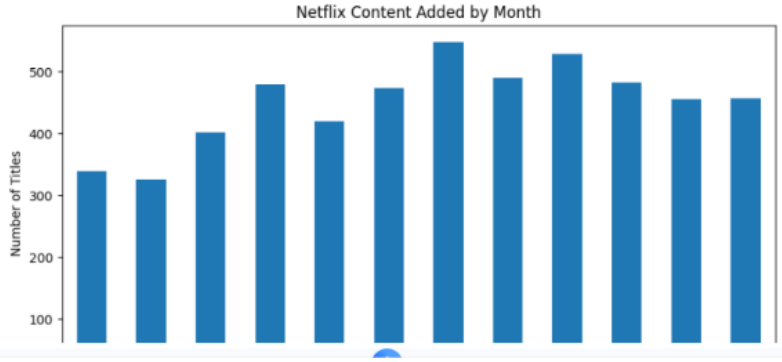
```
# Muthuraj R (241801176)
country_type = df[["country", "type"]].dropna().copy()
country_type["country"] = country_type["country"].str.split(",")
country_type = country_type.explode("country")
top_countries = country_type["country"].value_counts().head(10).index
data = country_type[country_type["country"].isin(top_countries)]
table = pd.crosstab(data["country"], data["type"])
print(table)
table.plot(kind="bar", figsize=(12, 6))
plt.title("Movies vs TV Shows by Country")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

```
type            Movie  TV Show
country
 Canada             66       30
 United States     196       65
Canada              83       51
France              86       51
India              578       46
Japan               67      144
South Korea         31      116
Spain               93       43
United Kingdom     166      132
United States     1107      632
```



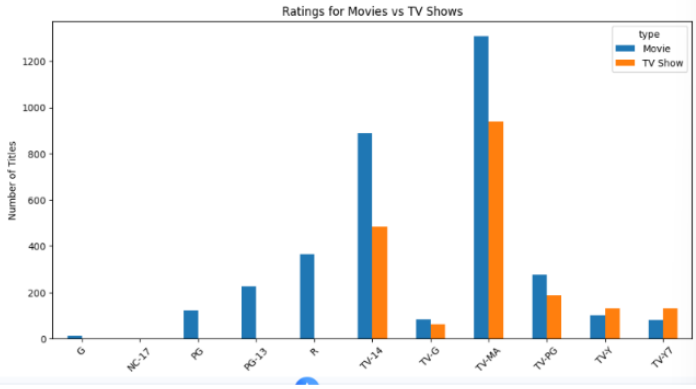

```
# STEP 8 MONTHLY CONTENT ADDITION
# Muthuraj R (241801176)
month_order = [
    "January", "February", "March", "April",
    "May", "June", "July", "August",
    "September", "October", "November", "December"
]
months = df["date_added"].dt.month_name()
monthly = months.value_counts().reindex(month_order)
print(monthly)
monthly.plot(kind="bar", figsize=(10, 5))
plt.title("Netflix Content Added by Month")
plt.xlabel("Month")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

```
date_added
January      338
February     325
March        402
April        479
May          420
June         473
July         548
August       490
September    528
October      482
November     455
December     457
Name: count, dtype: int64
```



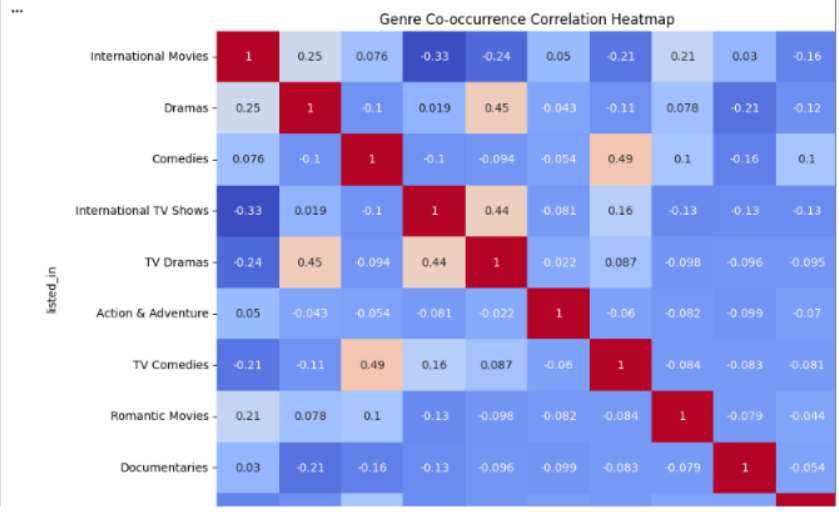
```python
# STEP 10 RATINGS BY CONTENT TYPE
# Muthuraj R (241801176)
rating_type = pd.crosstab(df["rating"], df["type"])
print(rating_type)
rating_type.plot(kind="bar", figsize=(12, 6))
plt.title("Ratings for Movies vs TV Shows")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```

```
type     Movie  TV Show
rating
G           13        0
NC-17        1        0
PG         123        0
PG-13      226        0
R          366        1
TV-14      888      485
TV-G        83       61
TV-MA     1308      938
TV-PG      277      188
TV-Y       100      130
TV-Y7       79      130
```




```python
# STEP 12 GENRE CORRELATION HEATMAP
# Muthuraj R (241801176)
genres = df["listed_in"].dropna().str.split(", ").explode()
genre_list = genres.value_counts().head(10).index
genre_data = pd.DataFrame(0, index=df.index, columns=genre_list)
for i, row in df["listed_in"].fillna("").items():
    for genre in genre_list:
        if genre in row:
            genre_data.loc[i, genre] = 1

correlation = genre_data.corr()
plt.figure(figsize=(12, 8))
sns.heatmap(correlation, annot=True, cmap="coolwarm")
plt.title("Genre Co-occurrence Correlation Heatmap")
plt.show()
```



RESULT:

The Netflix dataset was explored and visualized to examine content types, ratings, countries, release years, dates added, durations, TV show seasons, genres, and directors. A comparison of ratings by content type and a genre co-occurrence correlation heatmap were generated to identify patterns in the dataset.